

S7CAS-PLEIADES-CARTE-MERE

Pleiades

01

Revision 1.00

Date : 13/01/2026

Revision history	

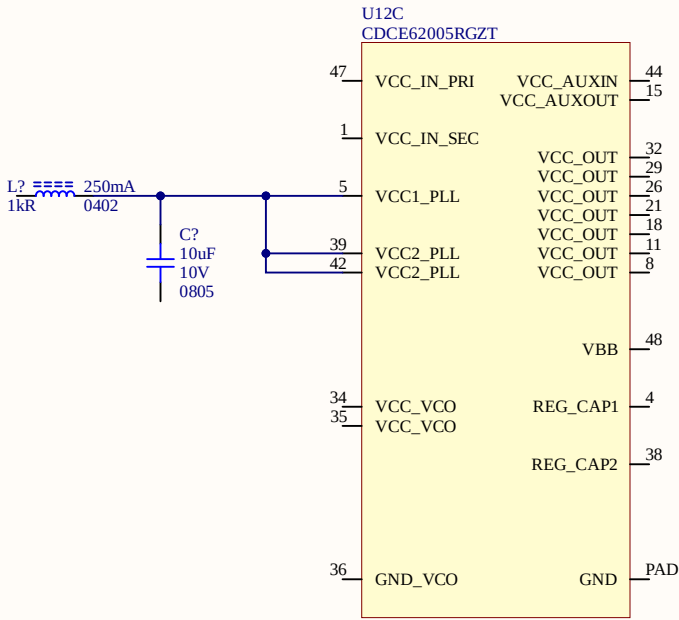
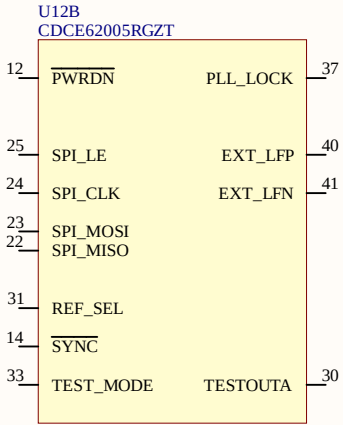
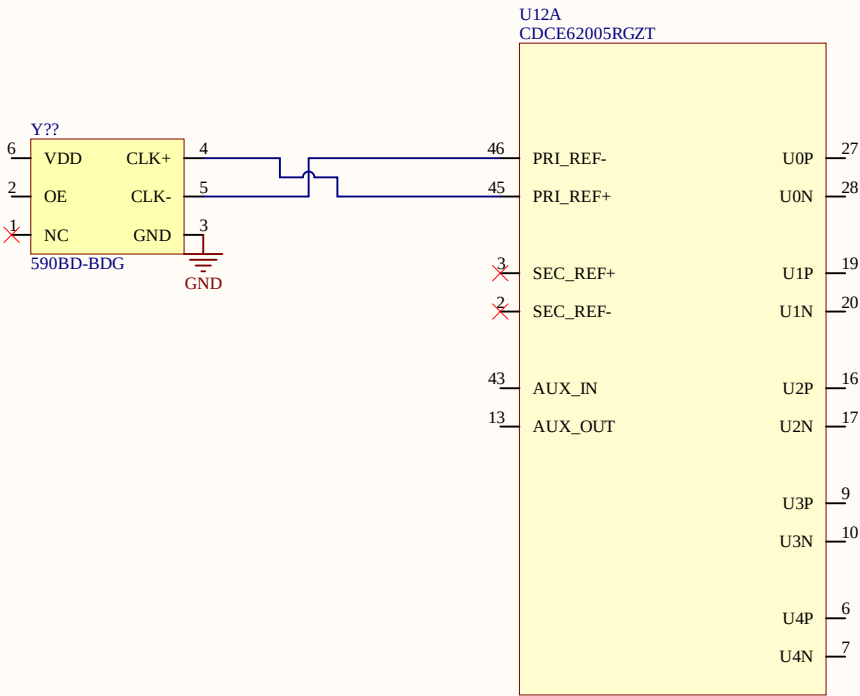
Package size conversion	
Metric	Imperial
1005	0402
1608	0603
2012	0805
3216	1206
3225	1210
6432	2512



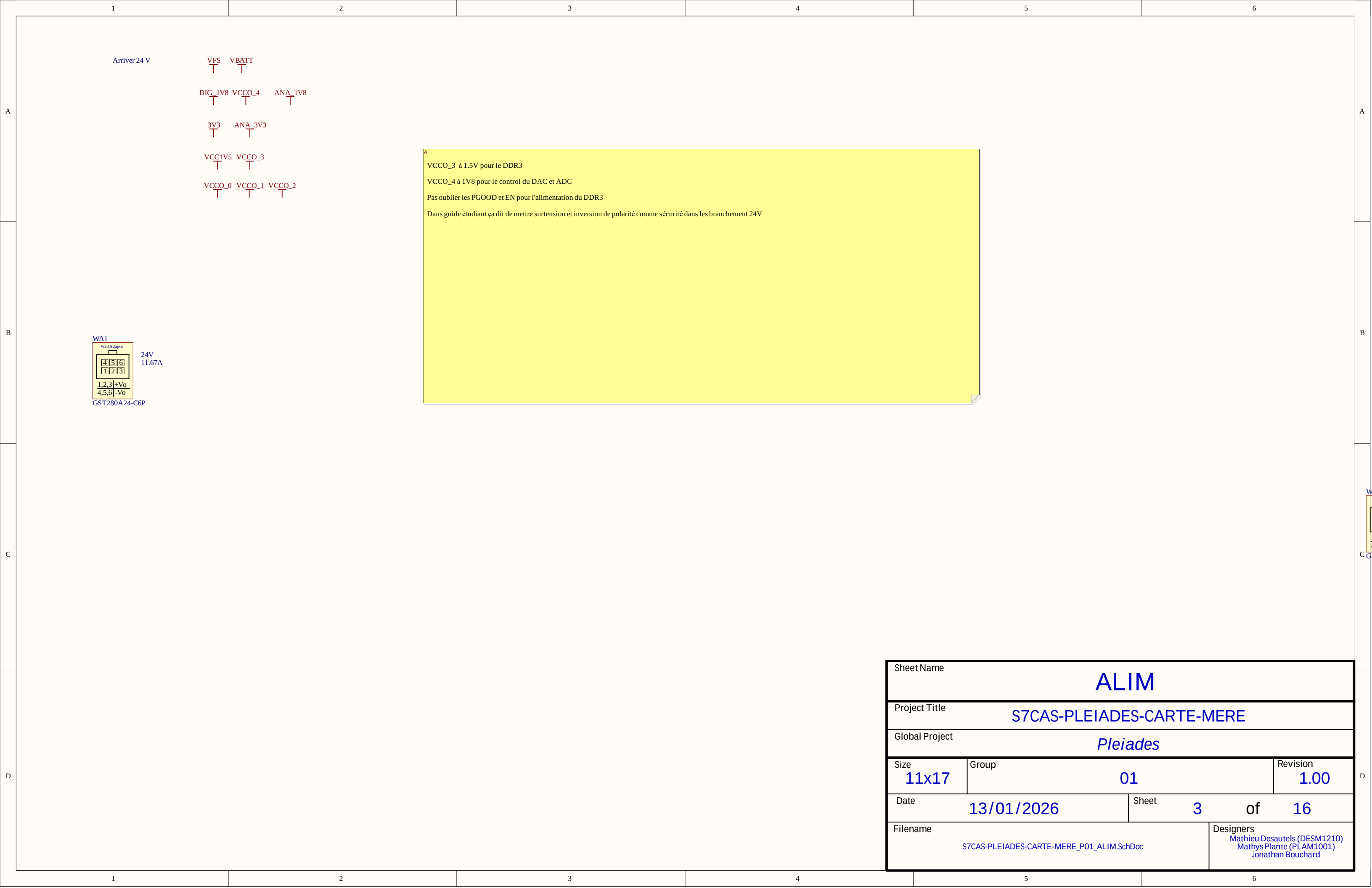
Project Title			S7CAS-PLEIADES-CARTE-MERE		
Global Project			Pleiades		
Size	Group		Revision		
11x17	01		1.00		
Date	13/01/2026		Sheet	1 of	16
Filename			Designers		
S7CAS-PLEIADES-CARTE-MERE_P01_TITLE.SchDoc			Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard		

CAN_CLK_P
CAN_CLK_N

CAN CLK 200MHZ
CNA CLK 400MHZ

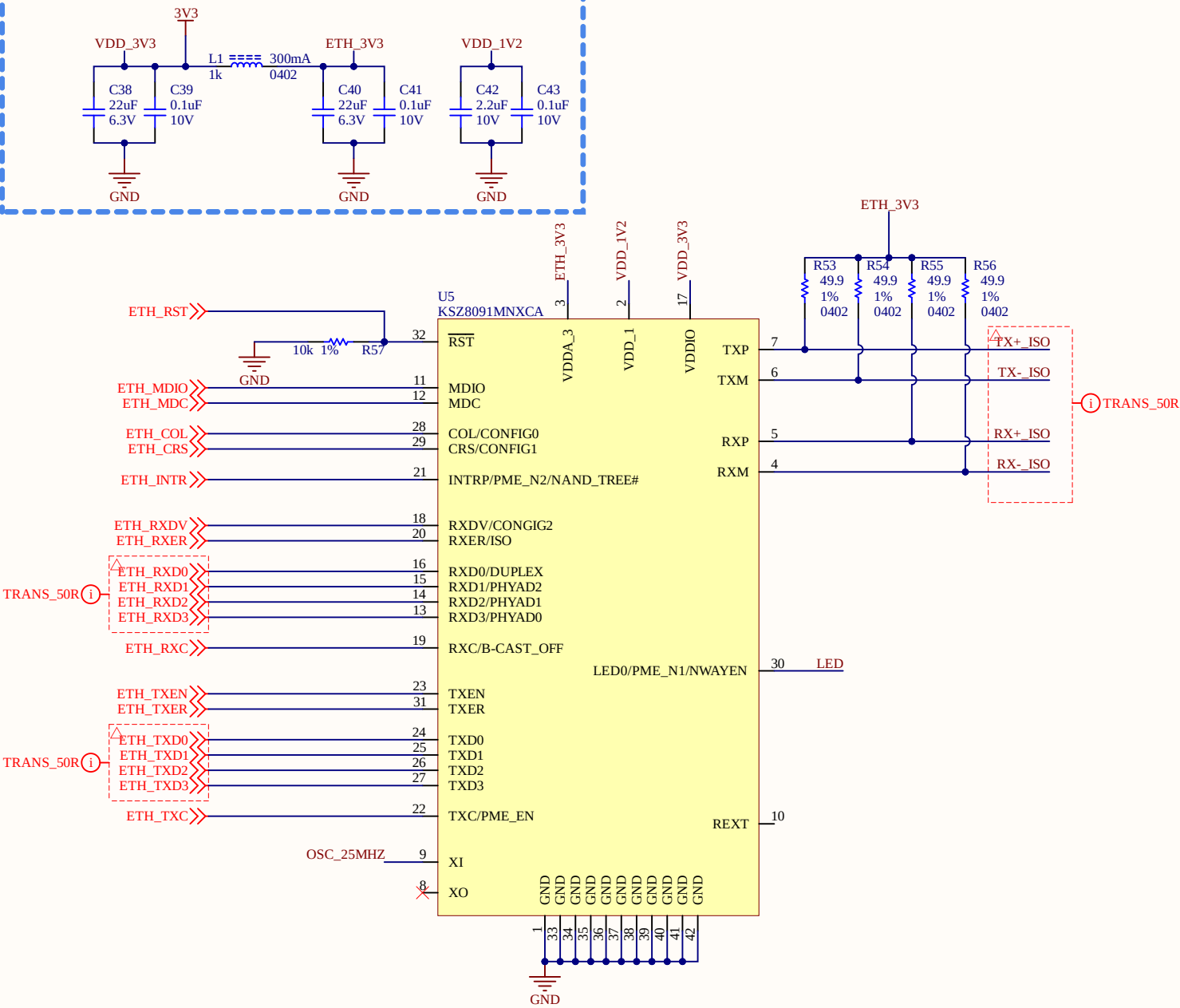


Sheet Name		CLOCK	
Project Title		S7CAS-PLEIADES-CARTE-MERE	
Global Project		Pleiades	
Size	Group	Revision	
11x17	01	1.00	
Date	13/01/2026	Sheet	2 of 16
Filename		Designers	
S7CAS-PLEIADES-CARTE-MERE_P01_CLOCK.SchDoc		Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard	



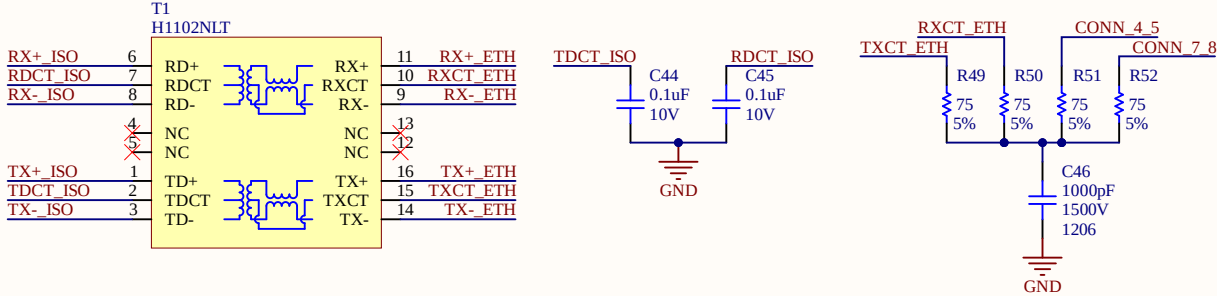
ETHERNET
ALIMENTATION

ETHERNET - PHY - KSZ8091MNXCA.pdf

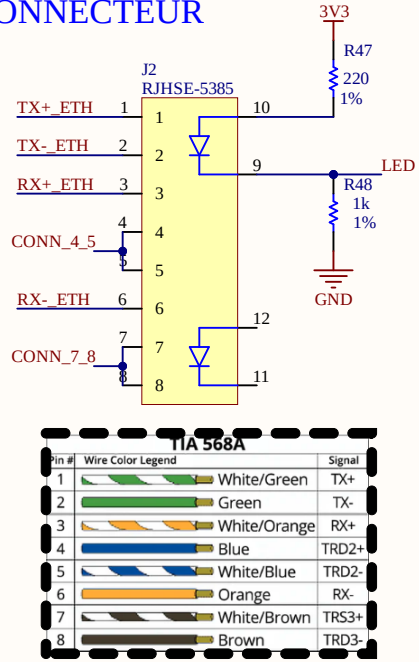


Ferite de 1k?
Valider les condensateur (Surtout pour ETH_3V3)
Faut tu mettre transmission entre Iso et Connecteur

ISO

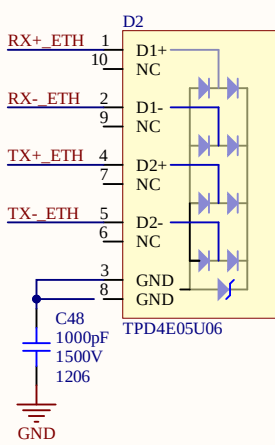


CONNECTEUR

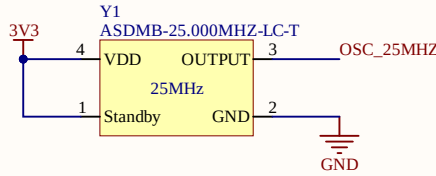


-Isolation du ETHERNET 1.5kV
-Ajout d'un condensateur de 1.5kV
-Ligne de transmission de 50R

ESD



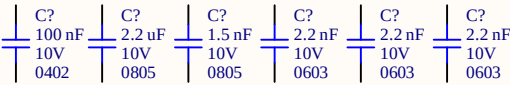
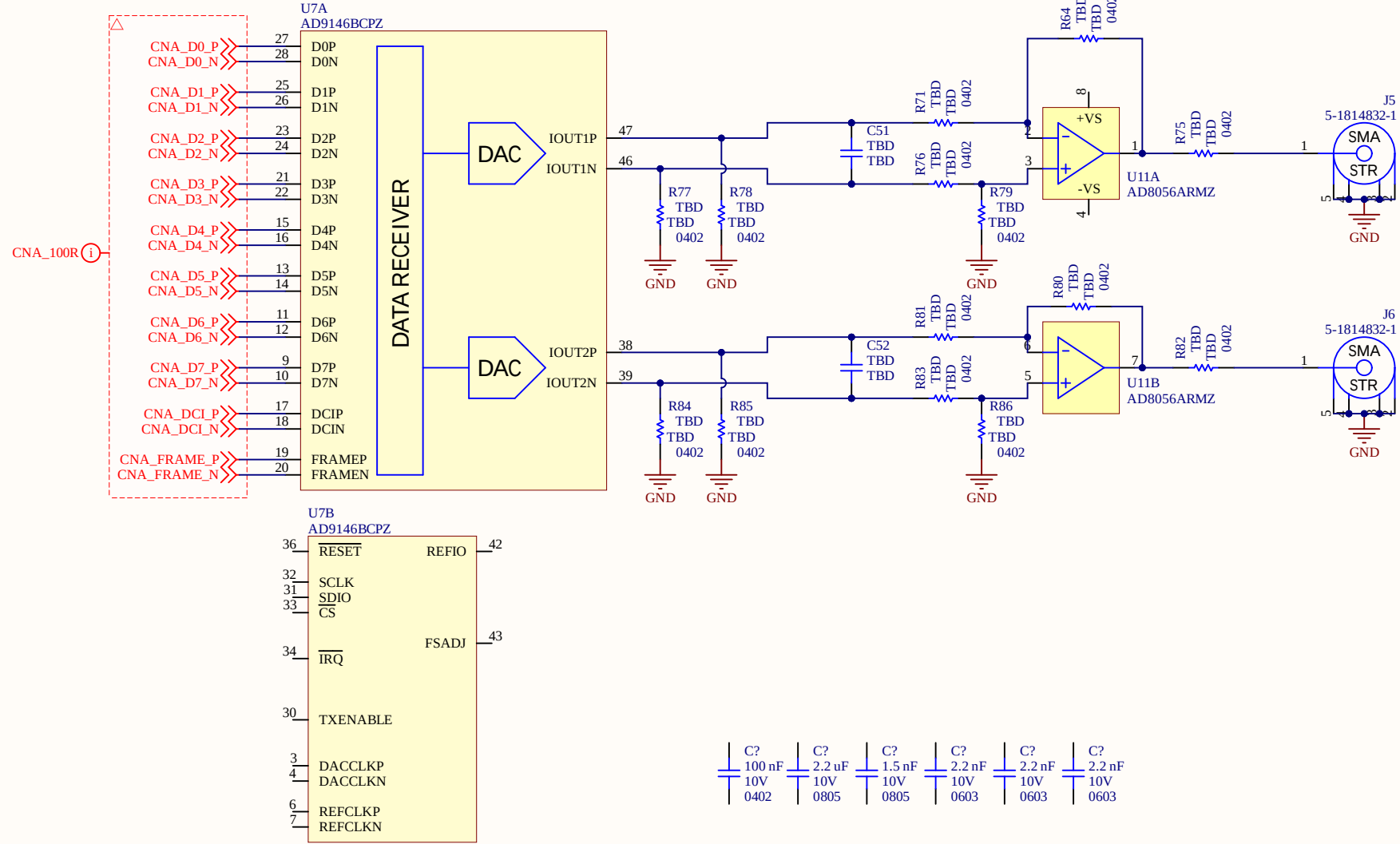
OSCILLI



Sheet Name		ETHERNET	
Project Title		S7CAS-PLEIADES-CARTE-MERE	
Global Project		Pleiades	
Size	Group	Revision	
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Date	13/01/2026	Sheet	4 of 16
Filename		Designers	
S7CAS-PLEIADES-CARTE-MERE_P01_ETHERNET.SchDoc		Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard	

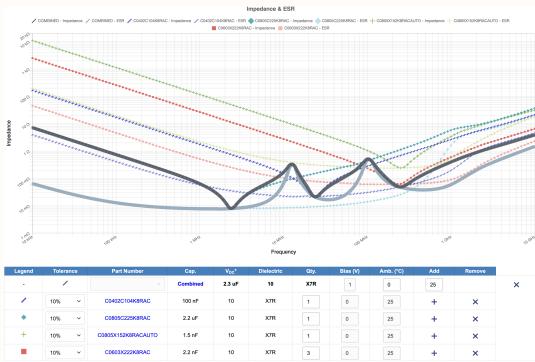
CNA (DAC)

Les clock c'est quel qu'on mets sur le FPGA et le quel qu'on mets sur l'oscillateur
- J'ai mit comme si la sortie c'est du courant
- Manque les inductances



200mbit/s en DDR LVDS

-Ajouter les condensateurs



Sheet Name			CNA		
Project Title			S7CAS-PLEIADES-CARTE-MERE		
Global Project			Pleiades		
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11x17	01			1.00	
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S7CAS-PLEIADES-CARTE-MERE_P01_CNA.SchDoc				Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard	

CAN (ADC)

DECOUPLAGE

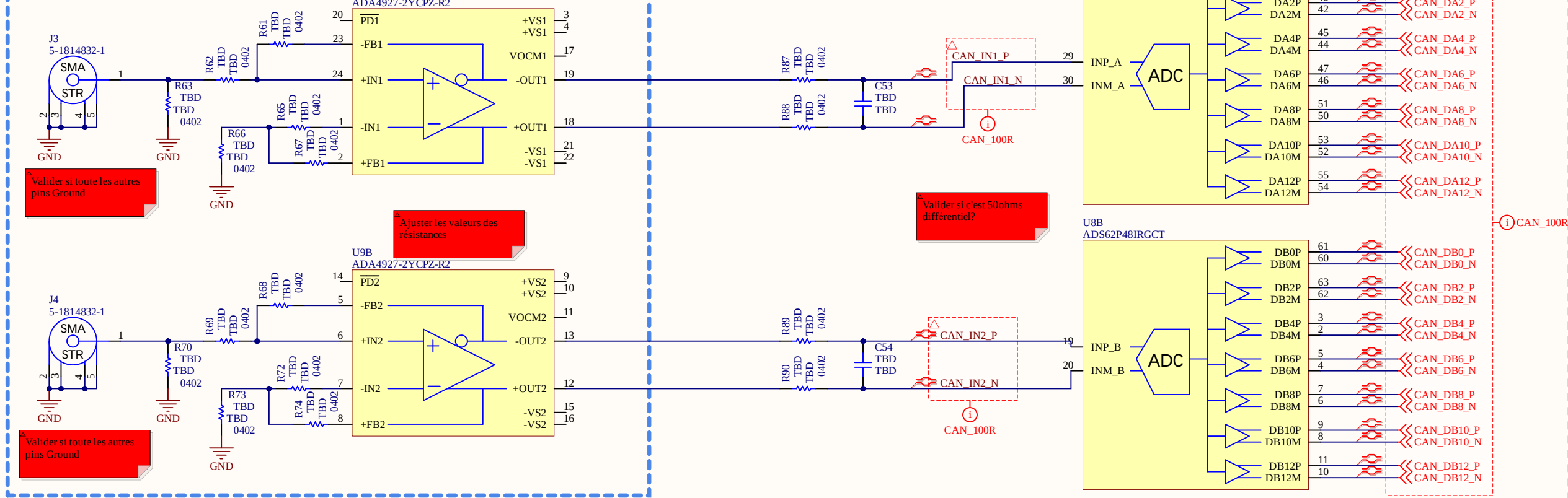
OPAMP - ADA4927-1_ADA4927-2.pdf
P.20

Ajuster les valeurs des résistances

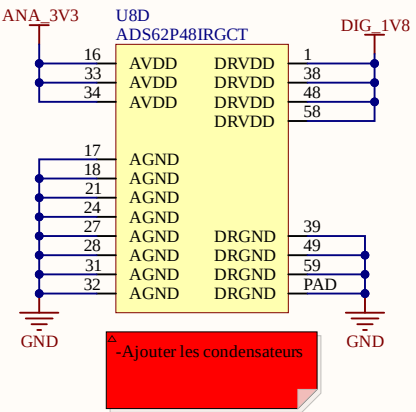
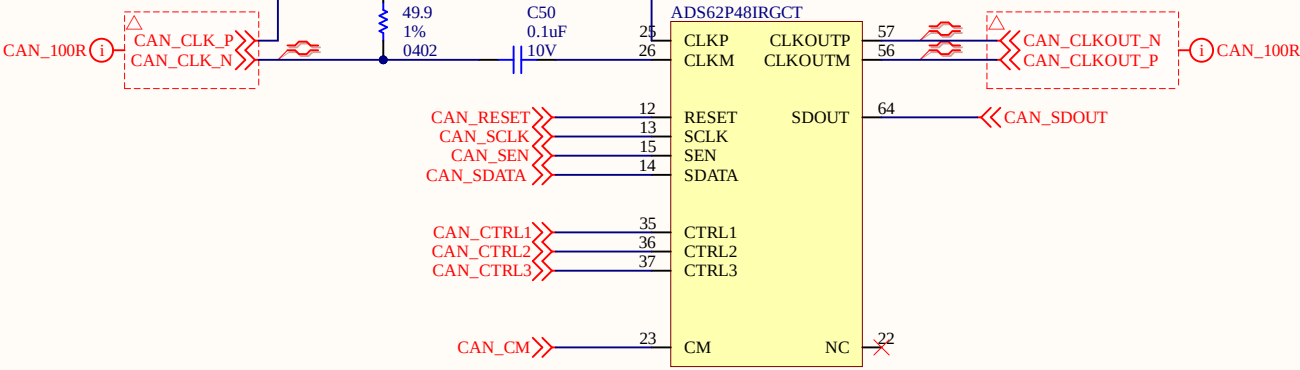
Manque les alims et inductances

Valider si c'est 50ohms différentiel?

-Choix de OP-AMP pour isolé et pour passer d'analog a digital



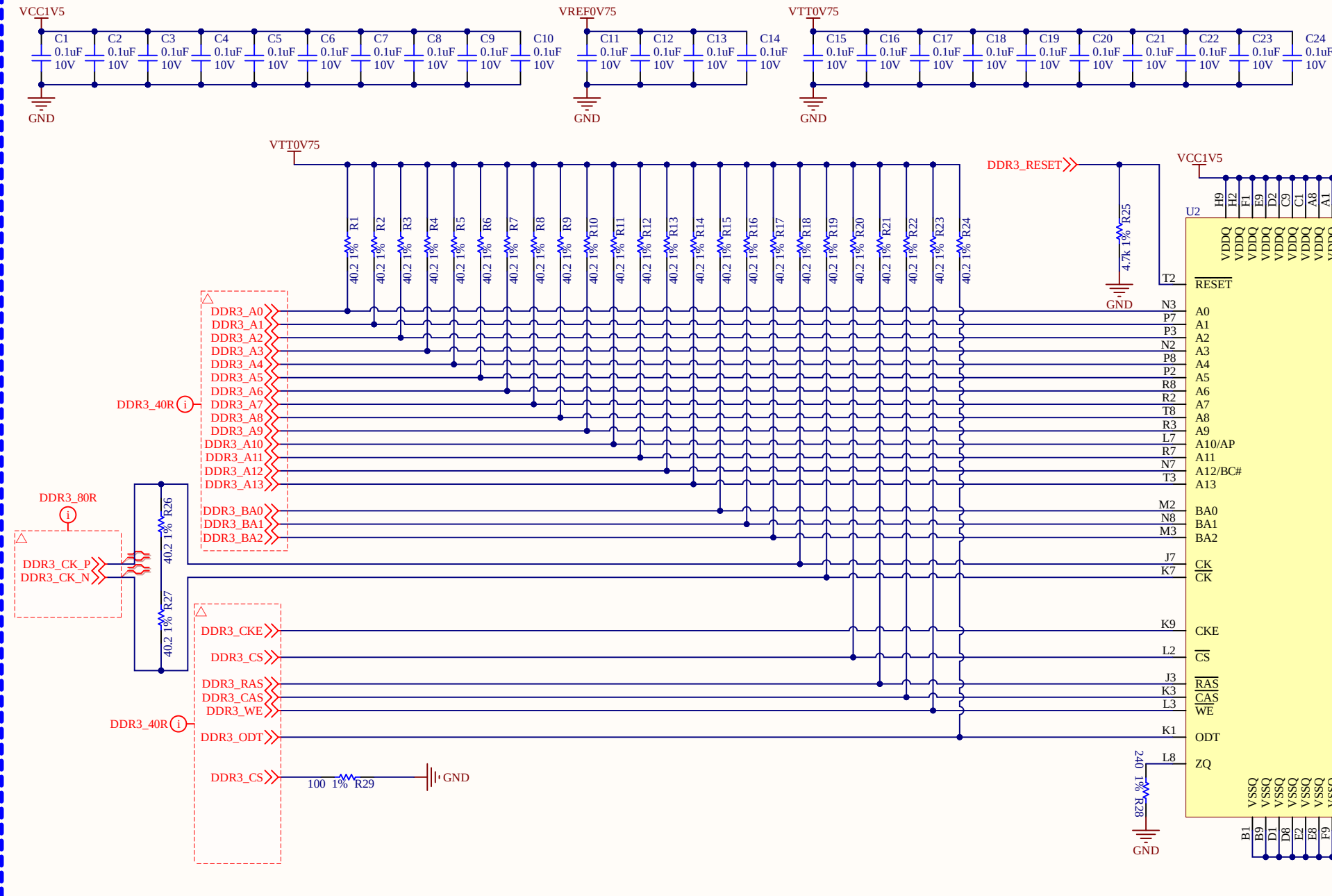
ADC - ADS62P48.pdf
Page 63



-Ajouter les condensateurs

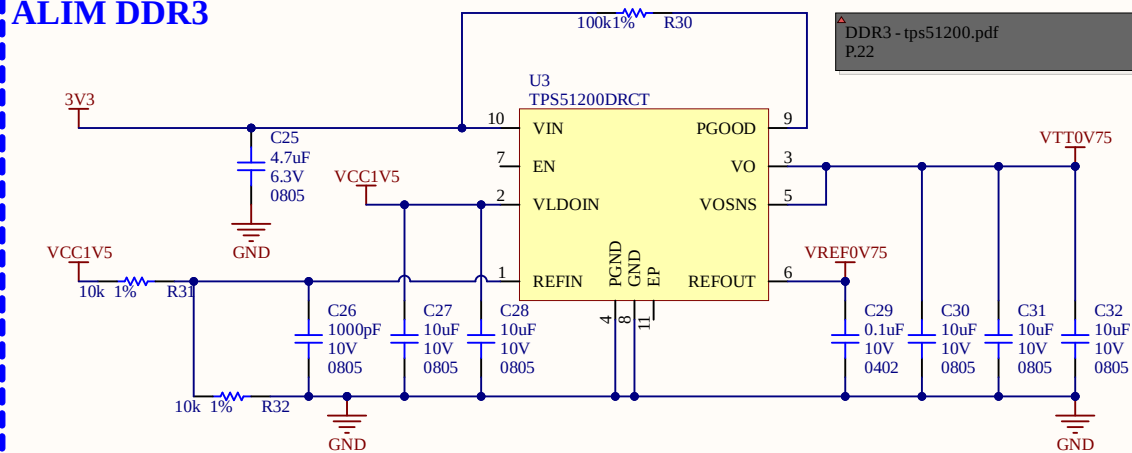
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Global Project			Pleiades		
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11x17	01			1.00	
Date	13/01/2026		Sheet	6	of 16
Filename			Designers		
S7CAS-PLEIADES-CARTE-MERE_P01_CAN.SchDoc			Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard		

FPGA



MANQUE ZIO?
ANALYSER LES COURANTS POUR LE VTT
AJUSTER LES CONDENSATEURS

ALIM DDR3

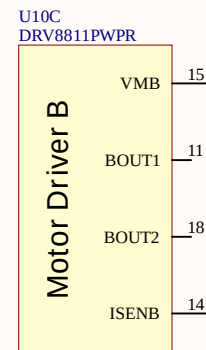
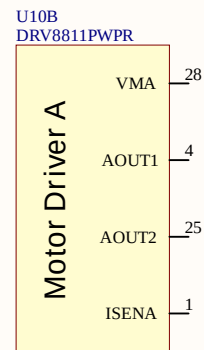
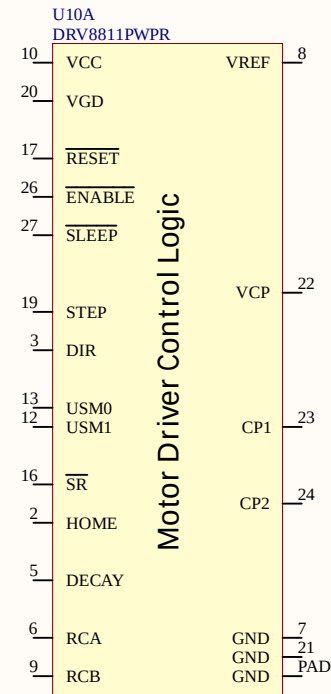


DDR3 - tps51200.pdf
P.22

-Ligne de transmission de 40R
-Resistance parallèle sur la ligne de transmission de 40.2Ohms
-Clock en pair différentiel 80R

Sortie PGOOD?
EN?

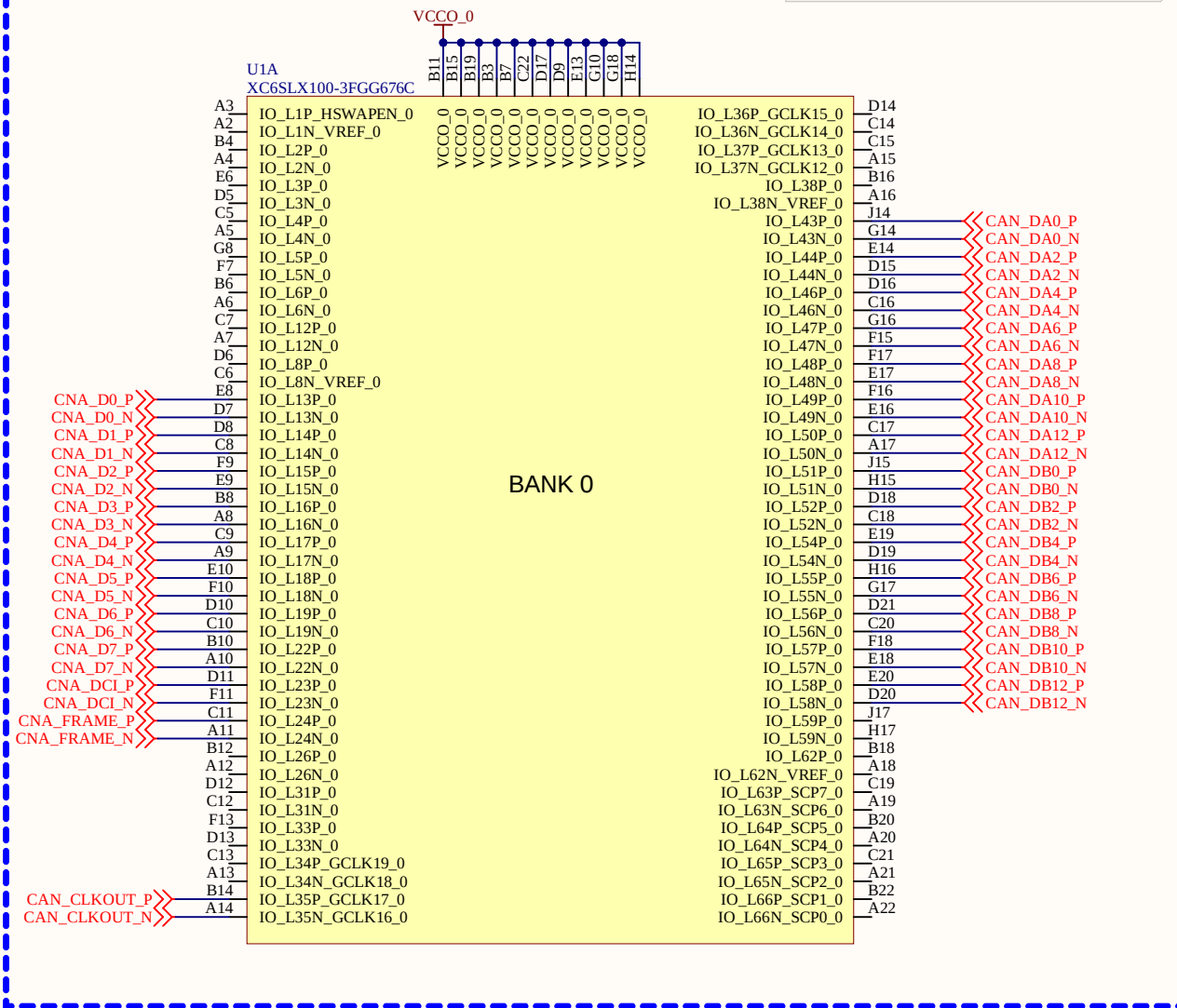
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Global Project			Pleiades		
Size	Group		Revision		
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Date	13/01/2026		Sheet	7	of 16
Filename			Designers		
S7CAS-PLEIADES-CARTE-MERE_P01_DDR3.SchDoc			Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard		



Sheet Name			MOTORS		
Project Title			S7CAS-PLEIADES-CARTE-MERE		
Global Project			Pleiades		
Size	Group	Revision			
11x17	01	1.00			
Date	13/01/2026	Sheet	8	of	16
Filename			Designers		
S7CAS-PLEIADES-CARTE-MERE_P01_MOTORS.SchDoc			Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard		

1	2	3	4	5	6
A					
B					
C					
D					

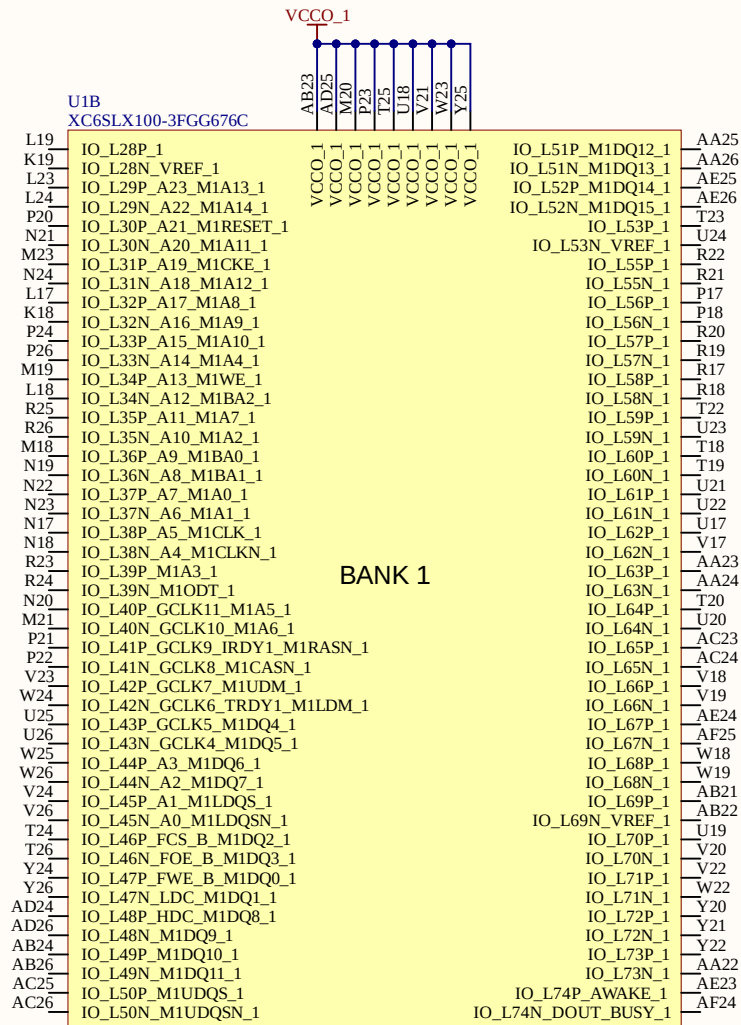
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Project Title		S7CAS-PLEIADES-CARTE-MERE	
Global Project		Pleiades	
Size	Group	Revision	
11x17	01	1.00	
Date	13/01/2026	Sheet	9 of 16
Filename		Designers	
S7CAS-PLEIADES-CARTE-MERE_P01_ASIC.SchDoc		Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard	



- LVDS seulement sur BANK0 et BANK2

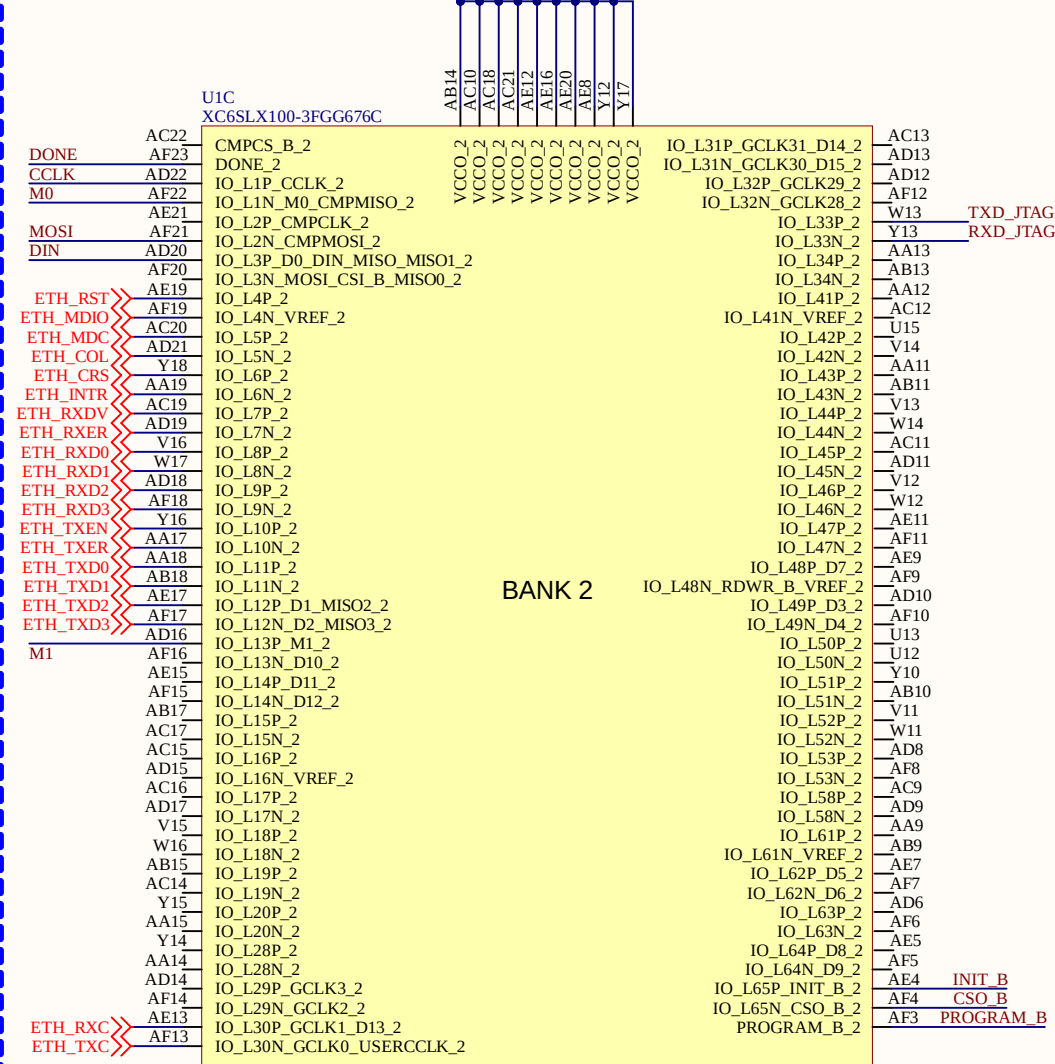
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Global Project			Pleiades		
Size	Group	Revision			
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Date	13/01/2026	Sheet	10	of	16
Filename			Designers		
S7CAS-PLEIADES-CARTE-MERE_P01_FPGA-BANK0.SchDoc			Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard		

FPGA

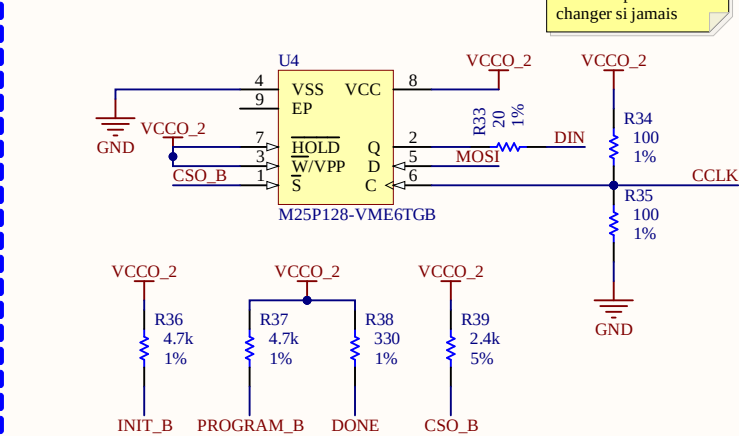


Sheet Name			FPGA - BANK 1		
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Global Project			Pleiades		
Size	Group	Revision			
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Date	13/01/2026	Sheet	11	of	16
Filename			Designers		
S7CAS-PLEIADES-CARTE-MERE_P01_FPGA-BANK1.SchDoc			Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard		

FPGA

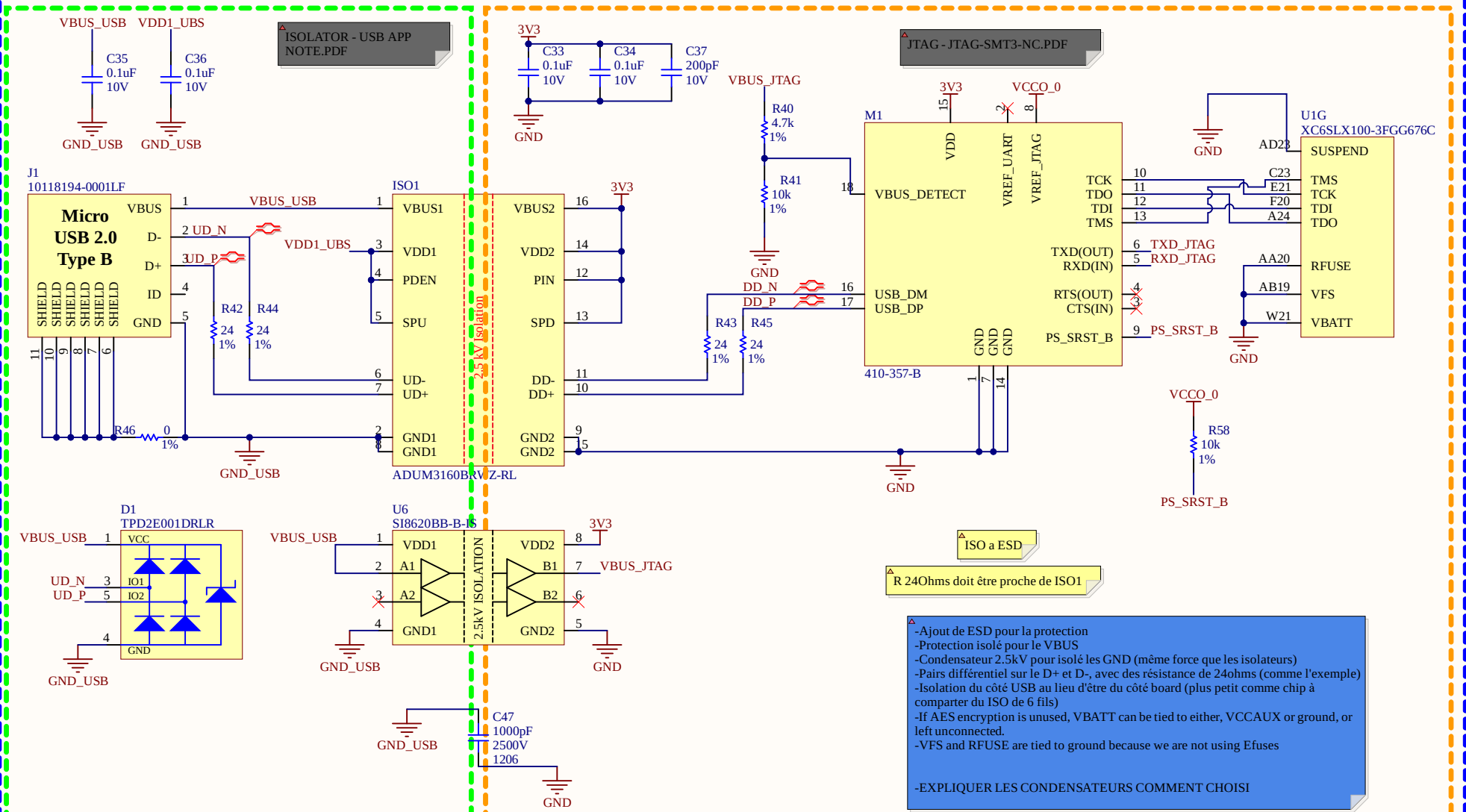


FLASH



SPARTAN6 - CONFIGURATION (UG380).PDF P42

USB

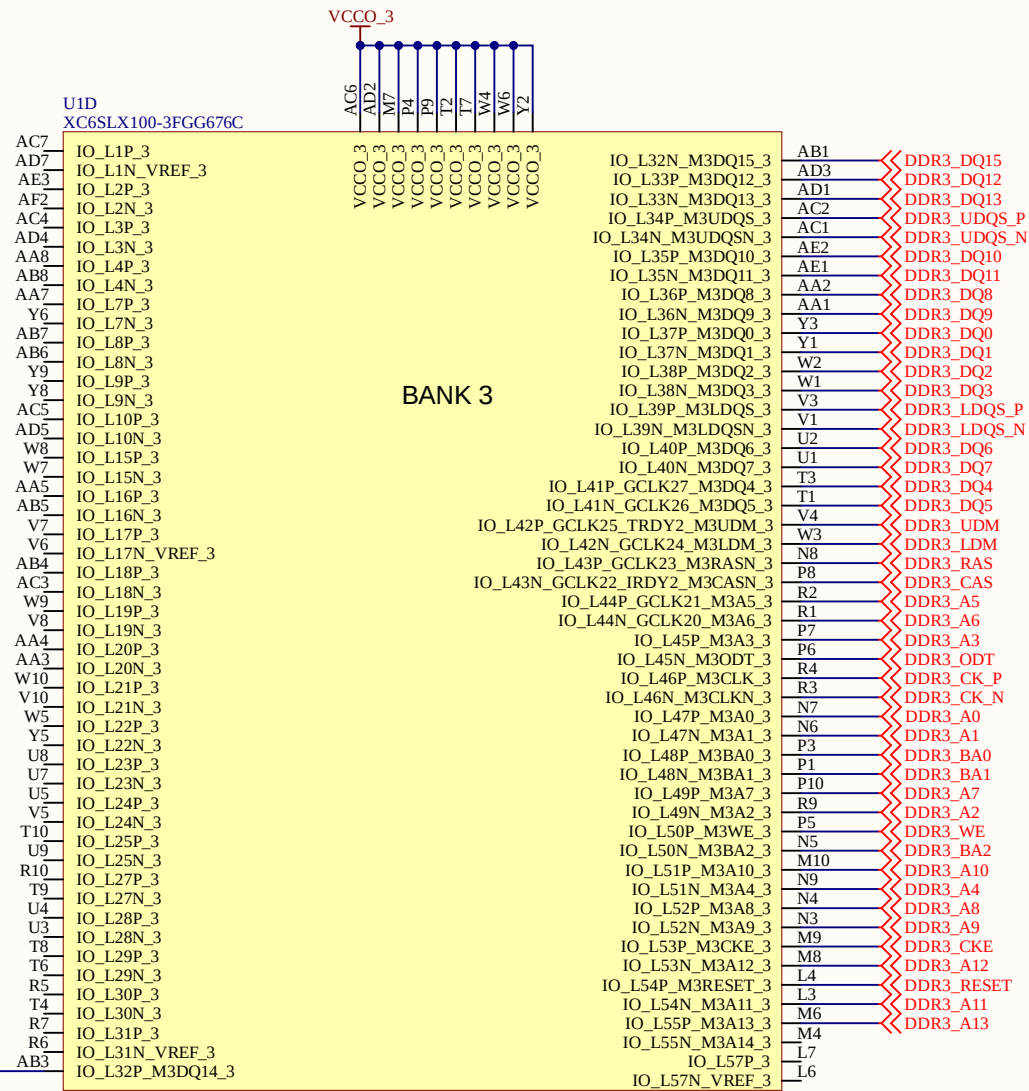


-Ajuster les condensateurs
-Pas sur si PS_SRST_B c'est obligatoire

-M0 et M1 choisi pour avoir le mode SPI
- Ajout d'une résistance de 20 ohms sur Q

Sheet Name			FPGA - BANK 2		
Project Title			S7CAS-PLEIADES-CARTE-MERE		
Global Project			Pleiades		
Size	Group			Revision	
11x17	01			1.00	
Date	13/01/2026		Sheet	12	of 16
Filename				Designers	
S7CAS-PLEIADES-CARTE-MERE_P01_FPGA-BANK2.SchDoc				Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard	

FPGA

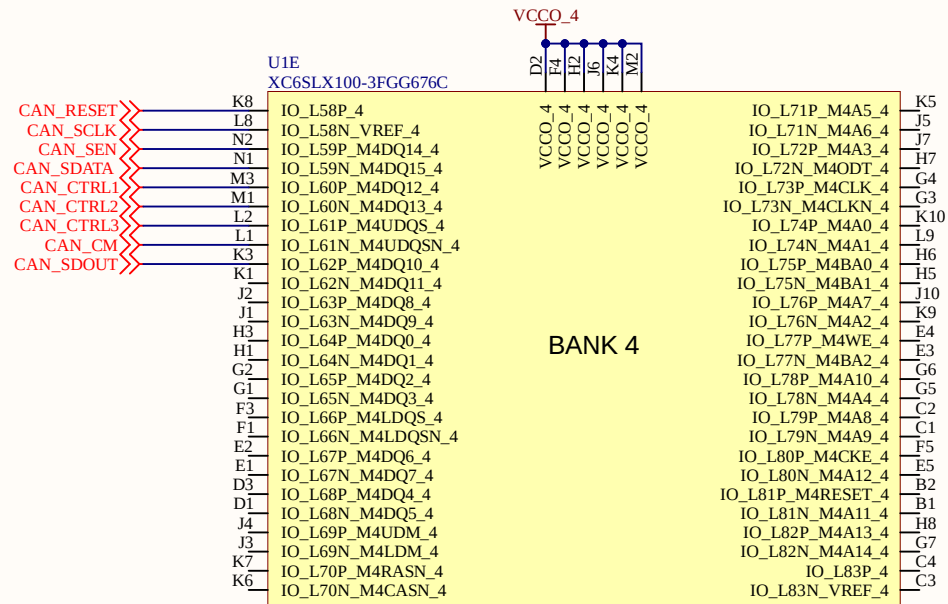


Rajouter les VREF avec des condensateurs pour le FPGA

VREF0V75

Sheet Name			FPGA - BANK 3		
Project Title			S7CAS-PLEIADES-CARTE-MERE		
Global Project			Pleiades		
Size	Group			Revision	
11x17	01			1.00	
Date	13/01/2026		Sheet	13	of 16
Filename				Designers	
S7CAS-PLEIADES-CARTE-MERE_P01_FPGA-BANK3.SchDoc				Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard	

FPGA



Sheet Name			FPGA - BANK 4		
Project Title			S7CAS-PLEIADES-CARTE-MERE		
Global Project			Pleiades		
Size	Group	Revision			
11x17	01	1.00			
Date	Sheet	of			
13/01/2026	14	16			
Filename			Designers		
S7CAS-PLEIADES-CARTE-MERE_P01_FPGA-BANK4.SchDoc			Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard		

